

Exploratory Data Analysis

Sprint 01 · Friday Session

Descriptive Statistics · Distributions · Correlation · Outlier Detection

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MicroMasters: Data Science, AI & Research Methods

How This Session Is Structured

Part A - Theory (this half)

1. Descriptive Statistics
2. Distributions & Visualisation
3. Correlation Analysis
4. Outlier Detection

We cover all concepts first, with examples drawn from child health data.

Part B - Practice (second half)

- Live coding in Python and R
- Dataset: **Nigeria NDHS 2024**
Children's Recode (KR file)
- One topic at a time, matching Part A

Saturday

You will apply the full pipeline independently.

Our Dataset: Nigeria NDHS 2024 - Children's Recode

The **Demographic and Health Survey (DHS)** is a nationally representative household survey conducted in over 90 countries.

The **Children's Recode (KR file)** contains one row per child under five years old and covers:

- Anthropometric measurements (HAZ, WAZ, WHZ)
- Stunting, wasting, and underweight status
- Vaccination and illness history
- Household wealth, region, maternal education

Key Variables We Will Use Today

Variable	Meaning
hw70	Height-for-age Z
hw71	Weight-for-age Z
hw72	Weight-for-height Z
v025	Urban / Rural
v024	Region
v190	Wealth index
v106	Mother's education
b4	Child sex

Descriptive Statistics

Theory: Part 1 of 4

What Are Descriptive Statistics?

Descriptive statistics **summarise** a dataset without making inferences about a wider population. They are always the first step in any analysis so never skip them.

Central Tendency

- Mean $\bar{x}$
- Median
- Mode

Spread

- Std deviation s
- Variance s^2
- IQR
- Range

Shape

- Skewness
- Kurtosis
- Quartiles
- Min / Max

Discussion

Height-for-age Z-score in the NDHS: mean = -1.4 , median = -1.6 . What does this gap tell us about the distribution of child stunting?

Mean vs. Median - Which One to Report?

Use the mean when:

- Distribution is roughly symmetric
- No strong outliers
- You need arithmetic operations downstream (e.g. regression)

Use the median when:

- Data is skewed (common in health data)
- Outliers are present
- Reporting household wealth, income, wait times

Skewness Rule of Thumb

Value	Interpretation
$ s < 0.5$	Symmetric
0.5–1.0	Moderately skewed
$ s > 1.0$	Highly skewed

Z-scores in the NDHS are often left-skewed, many children cluster near normal with a long tail toward severe malnutrition.

Standard Deviation

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

Interquartile Range

$$\text{IQR} = Q_3 - Q_1$$

Skewness (Fisher's definition)

$$g_1 = \frac{1}{n} \sum \left(\frac{x_i - \bar{x}}{s} \right)^3$$

Z-score standardisation

$$z_i = \frac{x_i - \bar{x}}{s}$$

(Note: HAZ/WAZ/WHZ in DHS are already Z-scores multiplied by 100.)

The Five-Number Summary



Why it matters for NDHS data:

- HAZ scores below -200 (i.e. -2 SD) indicate stunting
- WAZ scores below -200 indicate underweight
- WHZ scores below -200 indicate wasting

DHS Coding Note

In the KR file, Z-scores are stored as integers multiplied by 100. A value of -250 means -2.50 SD. Values of 9996–9999 are **missing codes** so always filter these out before analysis.

Distributions & Visualisation

Theory: Part 2 of 4

What Is a Distribution?

A **distribution** describes how values of a variable are spread across its range. It tells us:

- Where values are most concentrated
- How wide the spread is
- Whether the data is symmetric or skewed
- Whether unusually extreme values exist

The Normal Distribution is the benchmark. Many statistical methods assume it but public health data rarely is perfectly normal.

The 68–95–99.7 Rule

For a normal distribution:

- 68% of values lie within $\pm 1\sigma$
- 95% of values lie within $\pm 2\sigma$
- 99.7% of values lie within $\pm 3\sigma$

DHS Z-scores are *designed* to follow $\mathcal{N}(0, 1)$ in a healthy reference population. Deviations from this in our data are themselves a finding.

Choosing the Right Plot

Goal	Plot Type	Notes
Show distribution shape	Histogram + KDE	Always adjust bin count
Compare groups	Box plot	Good for ≥ 3 groups
Show distribution AND spread	Violin plot	Better than box plot alone
Check normality	Q-Q plot	Points should follow diagonal
Two continuous variables	Scatter plot	Add a regression line
Many pairs at once	Pair plot	Useful before modelling

Rule for Your Report

Every plot must have a title, labelled axes, and a one-sentence caption. An unlabelled figure has no scientific value.

Reading Distribution Shapes

Symmetric

Mean $\approx$ Median

Tails are equal

Example: height in a healthy reference population

Right-Skewed

Mean $>$ Median

Long right tail

Example: household income, disease duration

Left-Skewed

Mean $<$ Median

Long left tail

Example: HAZ scores in a malnourished population

What to Do with Skewed Data

- Report the **median and IQR** rather than mean and SD
- Consider a **log or square-root transformation** before modelling
- Use **Spearman** instead of Pearson for correlation (covered next)

When to Transform Data

Some variables in the NDHS are strongly right-skewed and benefit from transformation before analysis.

Log transformation $x' = \log(x)$

- Use for counts, rates, income
- Cannot apply to zero or negative values
- Back-transform results for interpretation

Square-root transformation $x' = \sqrt{x}$

- Milder than log
- Works for non-negative counts

Z-Scores Are Already Standardised

HAZ, WAZ, and WHZ are already on a standard scale. You generally do **not** need to transform them. What you *do* need to do is **recode missing values** (9996–9999) to NA before any analysis.

Correlation Analysis

Theory: Part 3 of 4

What Is Correlation?

Correlation measures the **strength and direction** of the relationship between two variables.

Pearson r

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{(n - 1) s_x s_y}$$

- Measures *linear* association
- Both variables should be continuous and roughly normal
- Sensitive to outliers

Spearman ρ

- Same formula applied to the *ranks* of x and y
- Measures monotonic (not just linear) association
- Robust to outliers
- Preferred for DHS health data

Both range from -1 (perfect negative) to $+1$ (perfect positive). Zero means no linear association, not no relationship.

Interpreting r or ρ

Absolute Value	Strength	Example in child health
0.9–1.0	Very strong	HAZ & WAZ in the same child
0.7–0.9	Strong	Wealth index & vaccination coverage
0.5–0.7	Moderate	Mother's education & skilled birth attendant
0.3–0.5	Weak	Facility density & stunting rate
0.0–0.3	Negligible	Child sex & wasting Z-score
Negative sign	Inverse	Poverty & immunisation rate

The Golden Rule: Correlation $\neq$ Causation

A correlation tells you two variables **move together**. It does not tell you why.

Classic example: Countries with more hospitals have higher recorded mortality rates. Do hospitals cause death? No sicker people go to hospitals.

NDHS example: Wealth index is strongly negatively correlated with stunting. Does wealth *cause* better nutrition? Possibly, but wealth also correlates with education, clean water, and healthcare access. Correlation cannot disentangle this.

Language to Use in Your Report

- ✓ *associated with*
- ✓ *correlated with*
- ✓ *linked to*
- ✓ *tends to increase/decrease with*
- ✗ *causes*
- ✗ *leads to*
- ✗ *because of*

Which Correlation Method to Use?

Situation	Method	Reason
Both variables continuous, symmetric	Pearson r	Assumes normality
One or both variables skewed	Spearman ρ	Rank-based, robust
Ordinal variables (wealth quintile, education)	Spearman ρ	No normality assumed
Outliers present	Spearman ρ	Ranks limit outlier influence
Small sample ($n < 30$)	Spearman ρ	Safer assumption

*For the Nigeria NDHS KR file: use **Spearman** as the default.*

Outlier Detection

Theory: Part 4 of 4

What Is an Outlier?

An outlier is an observation that lies **unusually far** from the rest of the data.

Types of outliers:

- **Data entry error:** Weight recorded as 450 kg
- **Measurement error:** Scale not calibrated
- **Genuine extreme observation:** A state with unusually high malnutrition
- **Valid but rare:** A very small or premature infant

The action you take depends entirely on which type it is. You must **investigate first**.

DHS Flagged Cases

The KR file contains a `hw73` variable: the DHS **flag** for implausible Z-scores. Any record where `hw73` $\neq$ 0 has been flagged by the DHS programme itself as biologically impossible. **Always exclude flagged records** before analysis.

The IQR Rule

The most common rule for flagging outliers in continuous data:

IQR Outlier Bounds

$$\text{Lower fence} = Q_1 - 1.5 \times \text{IQR} \quad \text{Upper fence} = Q_3 + 1.5 \times \text{IQR}$$

Any observation **below the lower fence or above the upper fence** is flagged.

Advantages

- Does not assume normality
- Based on the middle 50% of data
- What box plots use for their whiskers

Limitations

- May flag many observations in very skewed data
- The 1.5 multiplier is a convention, not a law
- Does not account for context

The Z-Score Rule

Standardise each observation and flag those that are more than 3 standard deviations from the mean:

$$z_i = \frac{x_i - \bar{x}}{s} \quad \text{flag if } |z_i| > 3$$

Advantages

- Intuitive: 99.7% of normal data lies within $\pm 3\sigma$
- Easy to compute

Limitations

- Assumes approximate normality
- In skewed data the mean and SD are already distorted by the outliers

Which Method for NDHS?

Use the **IQR rule** for general variables (wealth, age). Use the **DHS flag variable** (hw73) for anthropometric Z-scores, it is designed specifically for this purpose.

What to Do With an Outlier

Decision	When to use it	What to report
Keep	Genuine extreme observation	Document it; discuss in findings
Remove	Data or measurement error confirmed	State how many removed and why
Winsorise	Analysis requires all records	Cap at fence value; state the threshold
Transform	Right-skewed variable	Use log or sqrt; back-transform results

Do Not Auto-Delete

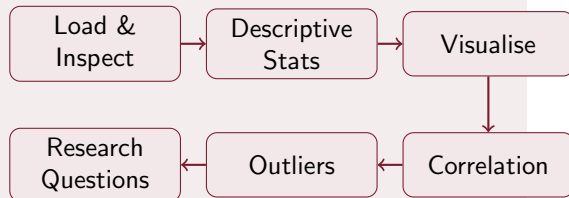
A child in the NDHS with a HAZ of -4.5 may represent severe stunting in a real, vulnerable household. Deleting that record hides the most important finding in your data.

Theory Summary - Before We Code

What we just covered:

1. **Descriptive stats**; mean, median, SD, skewness; choose the right measure for the data type
2. **Distributions**; histograms, box plots, violin plots; how to read shape and skewness
3. **Correlation**; Pearson vs. Spearman; how to interpret r or ρ ; never claim causation
4. **Outliers**; IQR rule, Z-score rule, DHS flag; investigate before deciding

The EDA Workflow



Practical Session

Nigeria NDHS 2024 - Children's Recode (KR file)

Setup - Loading the NDHS 2024 KR File

Python

```
import pandas as pd
import numpy as np
# Read the KR Stata file (.dta)
# pip install pyreadstat
df = pd.read_stata('NGKR8AFL.DTA')
# Or from CSV if already converted:
# df = pd.read_csv('NGKR8AFL.csv')
print(df.shape)
print(df.dtypes.head(20))
# Recode DHS missing values for Z-
# scores
# Values 9996-9999 are missing codes
for col in ['hw70', 'hw71', 'hw72']:
    df[col] = pd.to_numeric(df[col],
                           errors='coerce')
    df.loc[df[col] >= 9996, col] = np
    .nan
# Scale Z-scores (stored x100 in DHS)
df['haz'] = df['hw70'] / 100
df['waz'] = df['hw71'] / 100
df['whz'] = df['hw72'] / 100
```

R

```
library(tidyverse)
library(haven)
# Read Stata file
df <- read_dta("NGKR8AFL.DTA")
# Recode missing Z-score values
df <- df %>%
  mutate(
    haz = ifelse(hw70 >= 9996, NA,
                 hw70 / 100),
    waz = ifelse(hw71 >= 9996, NA,
                 hw71 / 100),
    whz = ifelse(hw72 >= 9996, NA,
                 hw72 / 100)
  )
# Remove DHS-flagged implausible
# records
# hw73 != 0 means flagged
df_clean <- df %>%
  filter(hw73 == 0 | is.na(hw73))
dim(df_clean)
```

Practical Session

Practice 1 - Descriptive Statistics

Practice 1 - Descriptive Statistics (Python)

```
# Summary statistics
print(df[['haz', 'waz', 'whz']].describe().T.round(3))

# Skewness and kurtosis
for col in ['haz', 'waz', 'whz']:
    print(f"{col}: {df[col].skew():.3f} {df[col].kurtosis():.3f}")

# Prevalence of malnutrition (binary indicators)
# hw70 < -200 = stunted; hw71 < -200 = underweight; hw72 < -200 = wasted
df['stunted'] = (df['haz'] < -2).astype(int)
df['underweight'] = (df['waz'] < -2).astype(int)
df['wasted'] = (df['whz'] < -2).astype(int)

print("\nMalnutrition Prevalence:")
print(df[['stunted', 'underweight', 'wasted']].mean().round(3) * 100)

# By region (v024) and residence (v025)
print(df.groupby('v024')['stunted'].mean().round(3) * 100)
```

Practice 1 - Descriptive Statistics (R)

```
library(skimr)
library(e1071)
# Rich summary
df_clean %>%
  select(haz, waz, whz) %>%
  skim()
# Skewness per outcome
sapply(df_clean[c('haz', 'waz', 'whz')],
       skewness, na.rm = TRUE)
# Define binary malnutrition indicators
df_clean <- df_clean %>%
  mutate(
    stunted      = as.integer(haz < -2),
    underweight  = as.integer(waz < -2),
    wasted       = as.integer(whz < -2))
# Prevalence by region
df_clean %>%
  group_by(v024) %>%
  summarise(
    stunting_pct = mean(stunted, na.rm=TRUE) * 100,
    wasting_pct  = mean(wasted,  na.rm=TRUE) * 100
  ) %>%
  arrange(desc(stunting_pct))
```

Practical Session

Practice 2 - Distributions & Visualisation

Practice 2 - Distributions (Python)

```
import matplotlib.pyplot as plt
import seaborn as sns

fig, axes = plt.subplots(1, 3, figsize=(14, 4))
labels = {'haz': 'Height-for-Age_Z_(HAZ)',
          'waz': 'Weight-for-Age_Z_(WAZ)',
          'whz': 'Weight-for-Height_Z_(WHZ)'}

for ax, col in zip(axes, ['haz', 'waz', 'whz']):
    sns.histplot(df[col].dropna(), bins=40,
                 kde=True, color='#7B1C2E',
                 line_kws={'lw': 2}, ax=ax)
    ax.axvline(-2, color='#C5A028', lw=1.5,
               linestyle='--', label='Cutoff_(-2_SD)')
    ax.set_title(labels[col])
    ax.set_xlabel('Z-Score')
    ax.legend(fontsize=8)

plt.suptitle('Distribution_of_Child_Anthropometric_Z-Scores\nNigeria_NDHS_2024',
             fontsize=12, y=1.02)
plt.tight_layout()
plt.savefig('fig1_zscore_distributions.png', dpi=150)
```

Practice 2 - Distributions (R)

```
library(ggplot2)
library(patchwork)

make_dist <- function(var, title) {
  ggplot(df_clean, aes(x = .data[[var]])) +
    geom_histogram(aes(y = after_stat(density)), bins = 40, fill = "#7B1C2E",
                  alpha = 0.75, na.rm = TRUE) +
    geom_density(color = "#C5A028",
                linewidth = 1, na.rm = TRUE) +
    geom_vline(xintercept = -2, color = "steelblue",
              linetype = "dashed", linewidth = 1) +
    labs(title = title, x = "Z-Score",
         caption = "Dashed line = -2 SD cutoff") +
    theme_minimal()}

p1 <- make_dist("haz", "Height-for-Age (HAZ)")
p2 <- make_dist("waz", "Weight-for-Age (WAZ)")
p3 <- make_dist("whz", "Weight-for-Height (WHZ)")

(p1 | p2 | p3) +
  plot_annotation(
    title = "Child Z-Score Distributions, Nigeria NDHS 2024")
ggsave("fig1_zscore_distributions.png", width = 14, height = 4, dpi = 150)
```


Practice 2 - Box Plots by Region (Python)

```
# Map region codes to labels if needed
region_labels = {1: 'North_Central', 2: 'North_East',
                  3: 'North_West', 4: 'South_East',
                  5: 'South_South', 6: 'South_West'}
df['region_lbl'] = df['v024'].map(region_labels)

fig, ax = plt.subplots(figsize=(10, 5))
sns.boxplot(data=df, x='region_lbl', y='haz',
            palette='Reds_r', ax=ax,
            flierprops=dict(marker='o',
                            markerfacecolor='gray',
                            markersize=2, alpha=0.4))
ax.axhline(-2, color='#C5A028', lw=1.5,
           linestyle='--', label='Stunting_cutoff')
ax.set_title('HAZ_Distribution_by_Region\nNigeria_NDHS_2024')
ax.set_xlabel('Region')
ax.set_ylabel('Height-for-Age_Z-Score')
ax.legend()
plt.tight_layout()
plt.savefig('fig2_haz_by_region.png', dpi=150)
```

Practical Session

Practice 3 - Correlation Analysis

Practice 3 - Correlation (Python)

```
import seaborn as sns
import matplotlib.pyplot as plt
# Select variables for correlation matrix
# v190 = wealth index, v106 = mother's education
# v025 = residence (1=urban, 2=rural)
corr_vars = df[['haz', 'waz', 'whz',
                'v190', 'v106', 'v025']].copy()
corr_vars.columns = ['HAZ', 'WAZ', 'WHZ', 'Wealth', 'MothersEduc', 'Residence']

# Compute Spearman correlation
corr = corr_vars.corr(method='spearman')

plt.figure(figsize=(7, 5))
sns.heatmap(corr, annot=True, fmt='.2f',
            cmap='RdYlGn', center=0,
            square=True, linewidths=0.4,
            cbar_kws={'shrink': 0.8})
plt.title('Spearman Correlation Matrix\nNigeria NDHS 2024 - Children Recode',
          pad=12)
plt.tight_layout()
plt.savefig('fig3_correlation_heatmap.png', dpi=150)
```

Practice 3 - Correlation (R)

```
library(corrplot)

corr_df <- df_clean %>%
  select(HAZ = haz, WAZ = waz, WHZ = whz,
         Wealth = v190, MothersEduc = v106,
         Residence = v025) %>%
  mutate(across(everything(), as.numeric)) %>%
  drop_na()

corr_matrix <- cor(corr_df, method = "spearman")

corrplot(corr_matrix,
         method      = "color",
         type        = "upper",
         addCoef.col  = "black",
         number.cex   = 0.8,
         tl.col       = "black",
         tl.cex       = 0.9,
         col = colorRampPalette(
           c("#7B1C2E", "white", "#1A6B3C"))(200),
         title = "Spearman Correlation: NDHS 2024",
         mar      = c(0, 0, 1.5, 0))
```

Practical Session

Practice 4 - Outlier Detection

Practice 4 - Outlier Detection (Python)

```
from scipy import stats

# Step 1: Remove DHS-flagged implausible records
df_valid = df[df['hw73'] == 0].copy()

# Step 2: IQR rule on HAZ
Q1 = df_valid['haz'].quantile(0.25)
Q3 = df_valid['haz'].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

iqr_flag = df_valid[(df_valid['haz'] < lower) |
                    (df_valid['haz'] > upper)]
print(f"IQR-flagged records: {len(iqr_flag)}")
print(f"Lower fence: {lower:.2f}")
print(f"Upper fence: {upper:.2f}")
print(iqr_flag[['haz', 'waz', 'v024', 'v025']].head(10))

# Step 3: Z-score rule
z = np.abs(stats.zscore(df_valid['haz'].dropna()))
print(f"\nZ-score flagged (|z|>3): {(z>3).sum()}")
```

Practice 4 - Outlier Detection (R)

```
# Step 1: Use only DHS-validated records
df_valid <- df_clean %>% filter(hw73 == 0)

# Step 2: IQR rule on HAZ
Q1 <- quantile(df_valid$haz, 0.25, na.rm = TRUE)
Q3 <- quantile(df_valid$haz, 0.75, na.rm = TRUE)
IQR_val <- Q3 - Q1

df_valid <- df_valid %>%
  mutate(iqr_flagged = haz < (Q1 - 1.5 * IQR_val) |
         haz > (Q3 + 1.5 * IQR_val))

cat("IQR-flagged records:",
    sum(df_valid$iqr_flagged, na.rm = TRUE), "\n")

# Visual: box plot highlighting flagged records
ggplot(df_valid, aes(x = "", y = haz)) +
  geom_boxplot(fill = "#7B1C2E", alpha = 0.6,
              outlier.colour = "orange",
              outlier.size = 1) +
  labs(title = "HAZ Distribution with IQR Outliers",
       x = NULL, y = "HAZ Score") +
  theme_minimal()
```

Key Takeaways from Today

1. **Describe before you model.** EDA reveals problems, patterns, and hypotheses that models alone cannot.
2. **Match your statistic to your data.** Skewed data → report median and IQR, use Spearman.
3. **DHS data needs specific preprocessing.** Recode missing Z-score values; exclude `hw73` $\neq$ 0 records before any analysis.
4. **Correlation $\neq$ causation.** Always use careful, hedged language in your report.
5. **Outliers are signals.** Investigate using the DHS flag first, then IQR. Never auto-delete.

For Saturday

Full independent EDA on the Nigeria NDHS 2024 KR file. Deliver a 2-page narrative report using the provided template. Use Python or R, your choice.

Questions?

See you Saturday for the full EDA application session.

ADA Global Academy · MicroMasters Sprint 01